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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
   - Profile setup: Create a user account (e.g., server / server123)
   - Install OpenSSH server: **Check this option** (Important!)
   - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode
4. Click **OK**

Install/Verify SSH Server

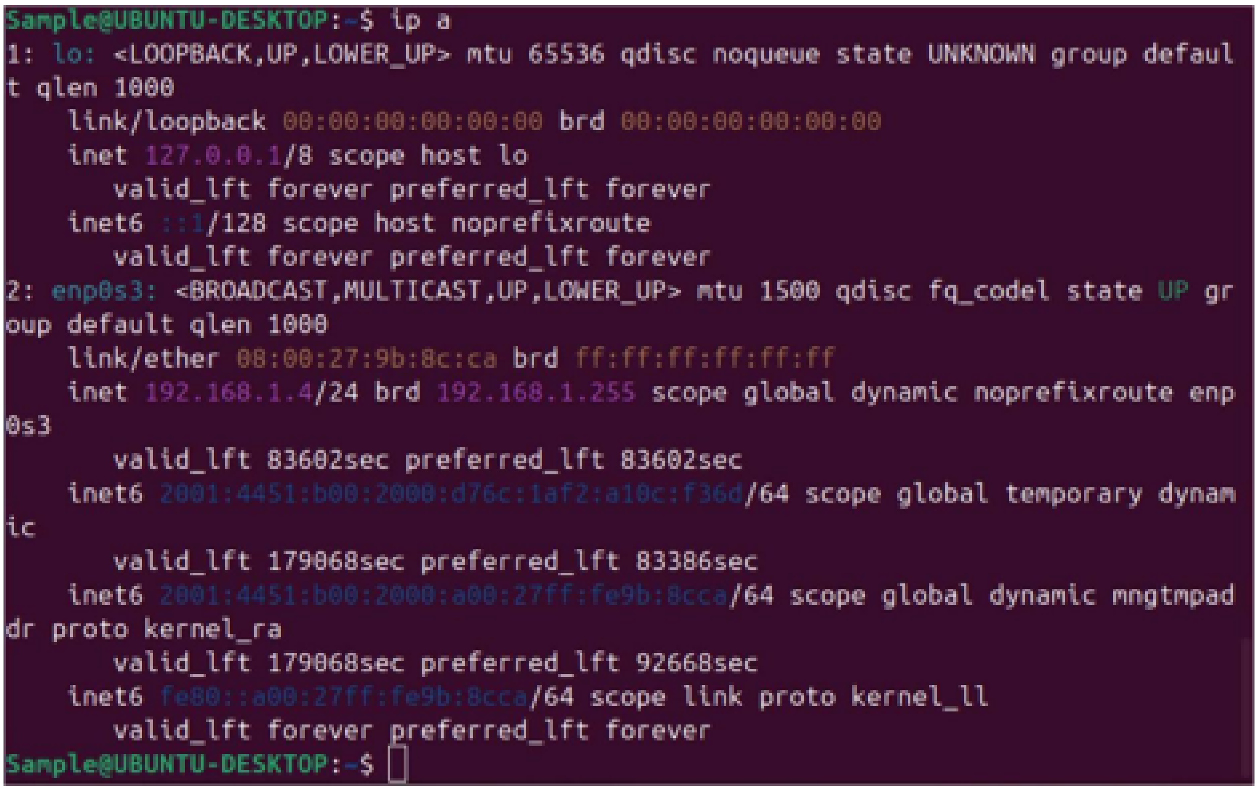
Server: Install SSH on the server (if not already installed):

- sudo systemctl status ssh – for checking the SSH status.
- sudo apt update – updating the your server.
- sudo apt install openssh-server -y –installing the SHH.
- sudo systemctl enable ssh – enabling the SSH to your unit. (SERVER/DESKTOP)
- sudo systemctlstartssh – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:



From the desktop, connect to the server: ssh username@192.168.1.120

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Sep  8 11:24:38 PM UTC 2025

  System load:            0.0
  Usage of /:             5.1% of 48.91GB
  Memory usage:           5%
  Swap usage:             0%
  Processes:              139
  Users logged in:        1
  IPv4 address for enp0s3: 192.168.1.14
  IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a


Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. **Screenshots (labeled clearly):**
   - VM network settings showing Bridged Adapter
   - Server IP address (output of ip a or hostname -I)
   - SSH service status (sudo systemctl status ssh)
   - Successful SSH connection from host
   - Verification commands executed inside VM

2. **Submission Format:**
   - Single PDF file
   - Include name, student ID, and date
   - All screenshots with brief explanation
   - Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

Name: ___________________________________
Student ID: ______________________________
Course/Year/Section: ______________________
Date Submitted: ___________________________
Instructor: _______________________________

1 OBJECTIVES (5 POINTS)

Check all that apply:

- □ **Create and configure a Virtual Machine with Ubuntu Server**
- □ **Configure Bridged Network Adapter for network access**
- □ **Determine and verify the server's IP address**
- □ **Install OpenSSH Server and enable the service**
- □ **Establish successful SSH connection from host machine**
- □ **Execute verification commands to confirm remote access**

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu (64bit)
Virtualization Software (VirtualBox/VMware)	Virtualbox
Ubuntu Server Version	Ubuntu 64bit
VM Name	Almendral-VM
RAM Allocated	8192RAM
Storage Allocated	100GB
Network Mode	NAT
SSH Client Used	OpenBSD Secure Shell

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	16MB
Hard Disk	100GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

Server IP Address: ______________________

Network Interface: ______________________

6. REMOTE CONNECTION

Connection Details:

Parameter	
Server IP	
Username	
SSH Port	

Value

169.254.81.182

almendral

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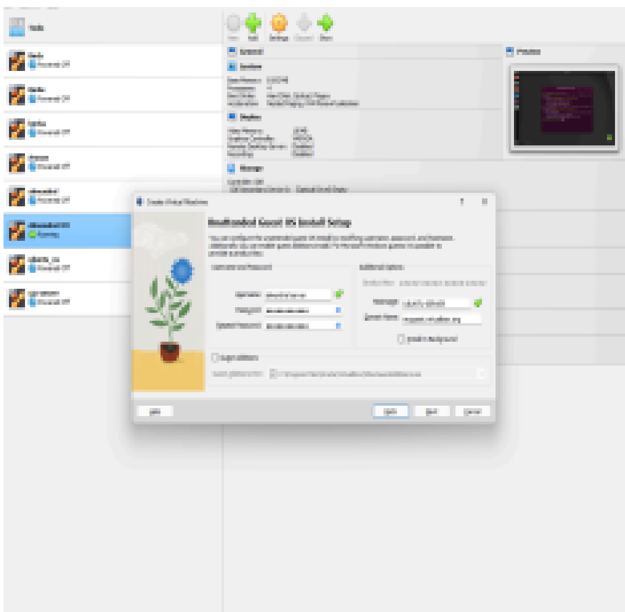
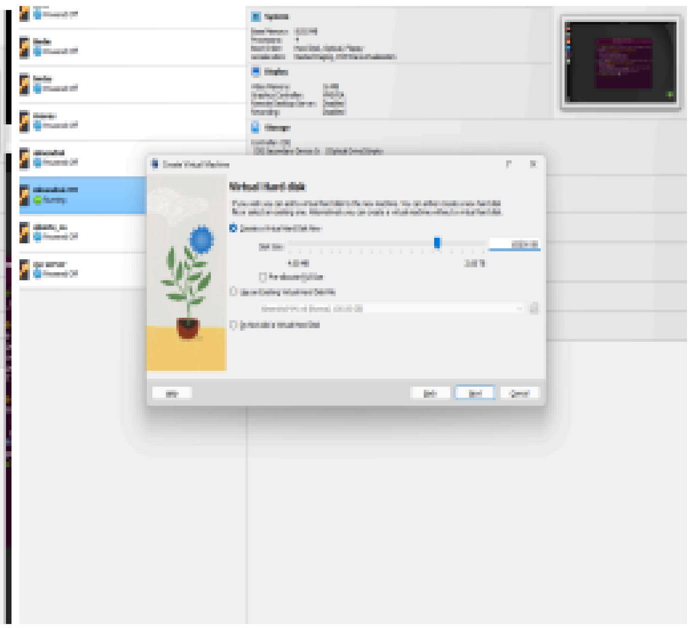
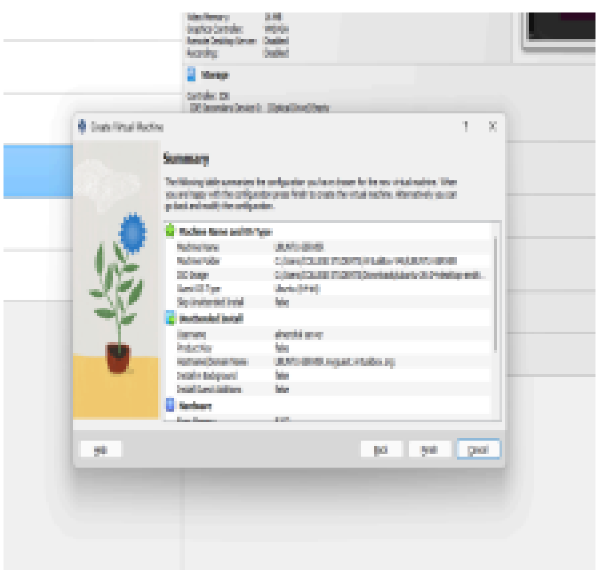
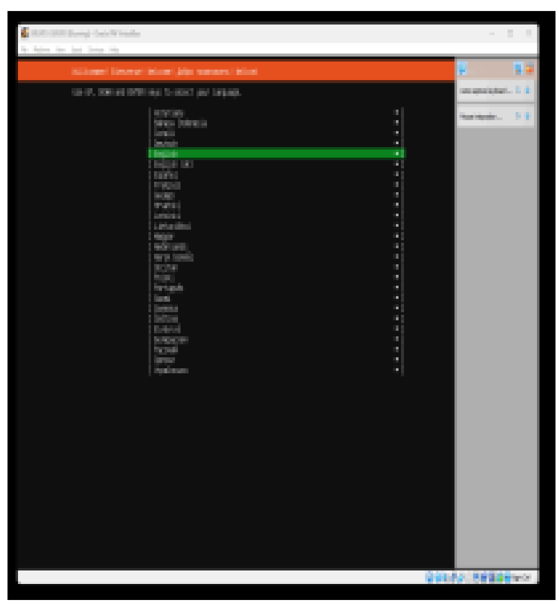
open a oracle vm and create a file for the ubuntu server

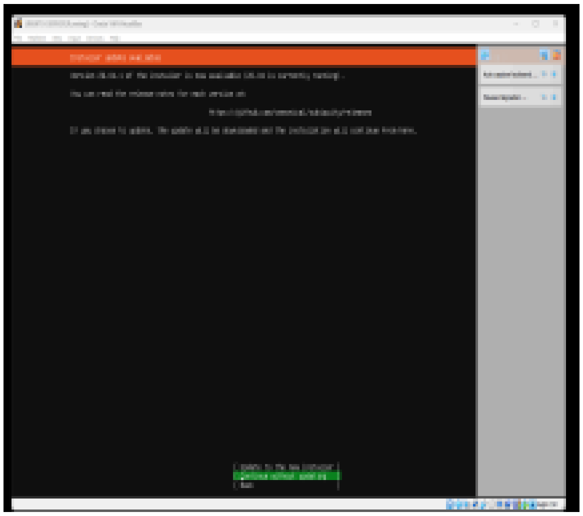


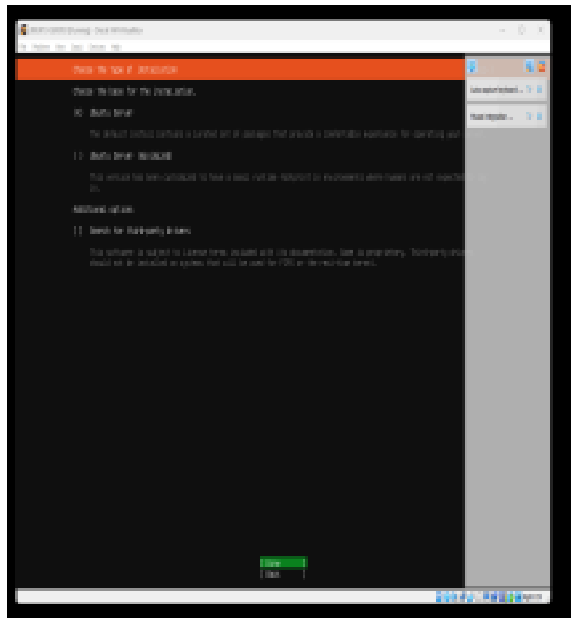


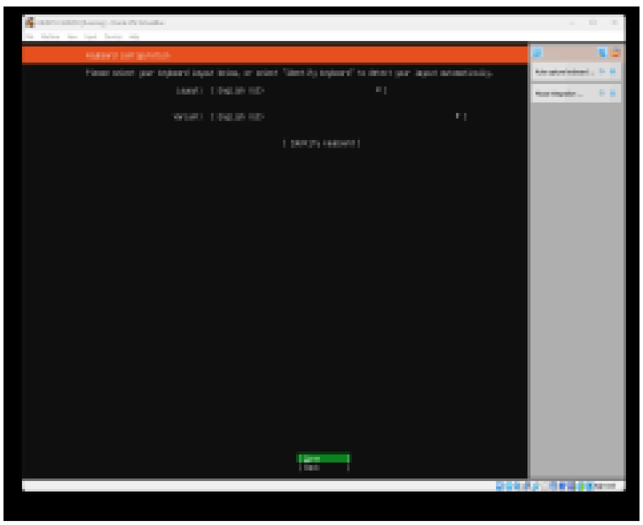
Finish and click a done

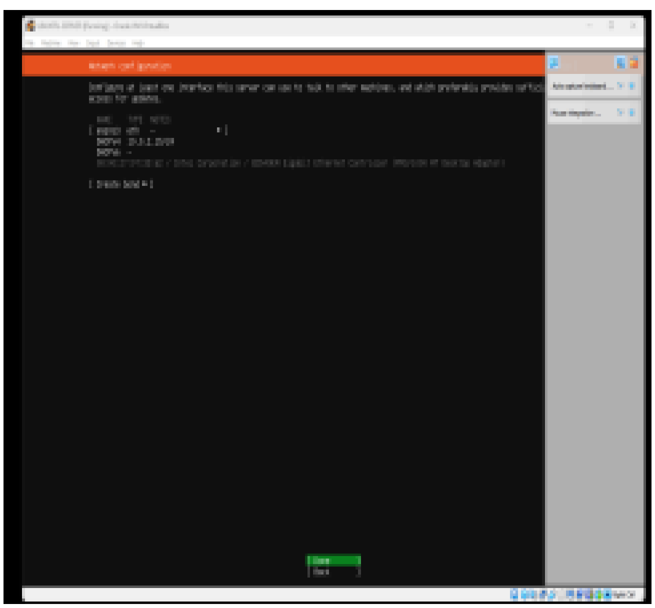


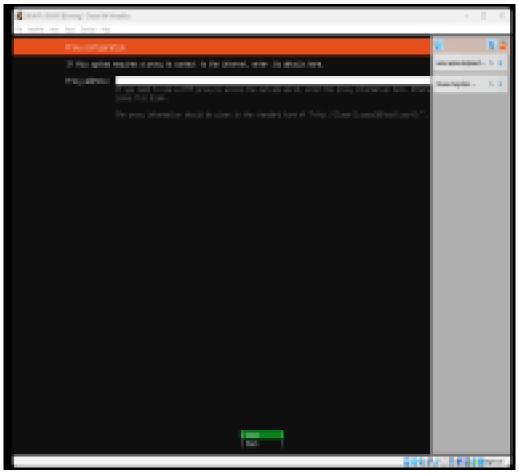


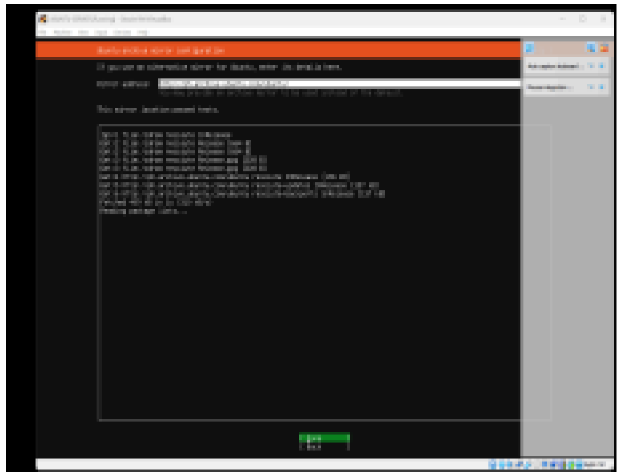




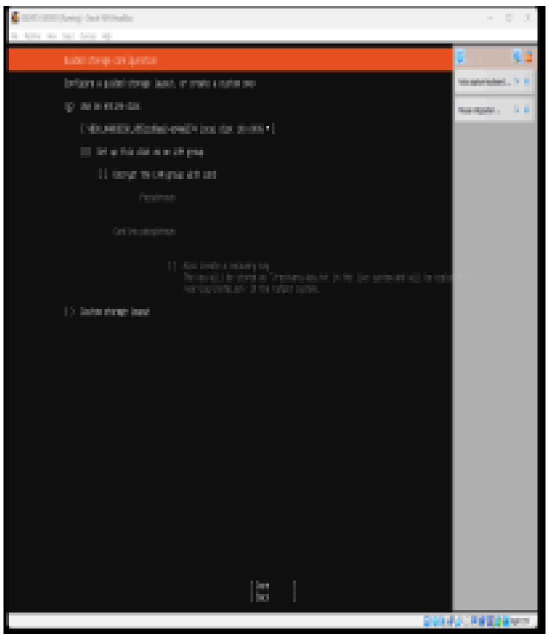


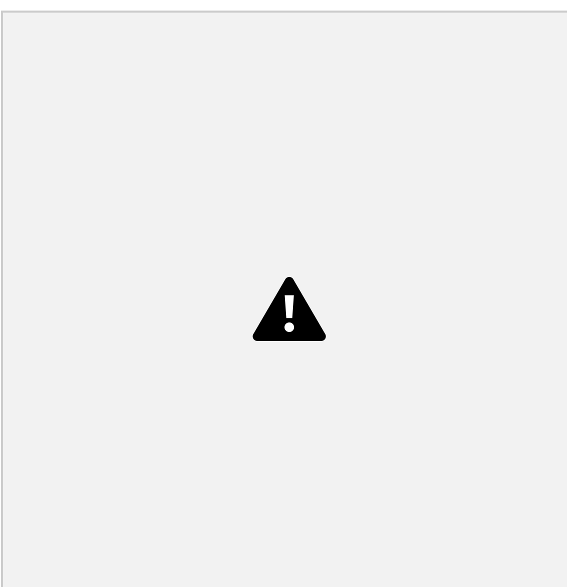


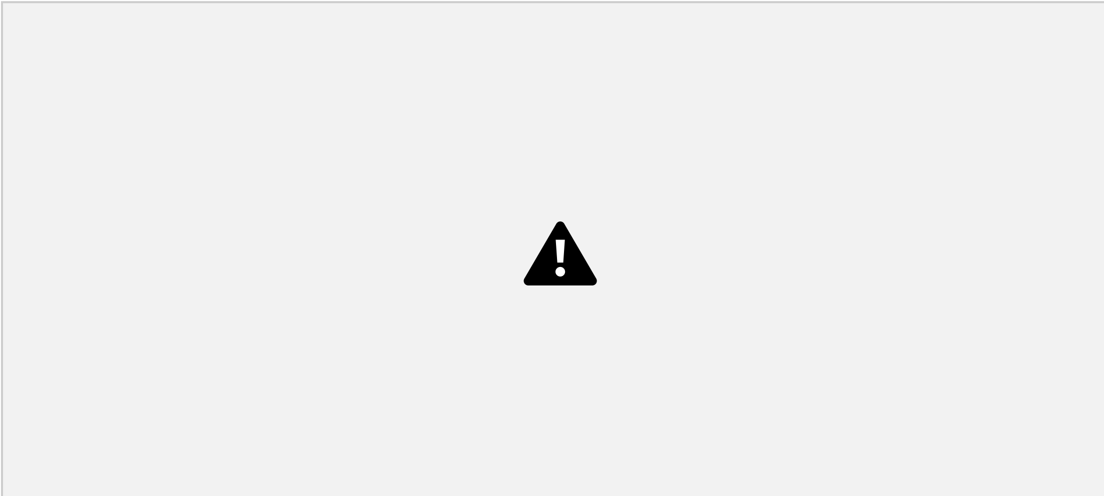




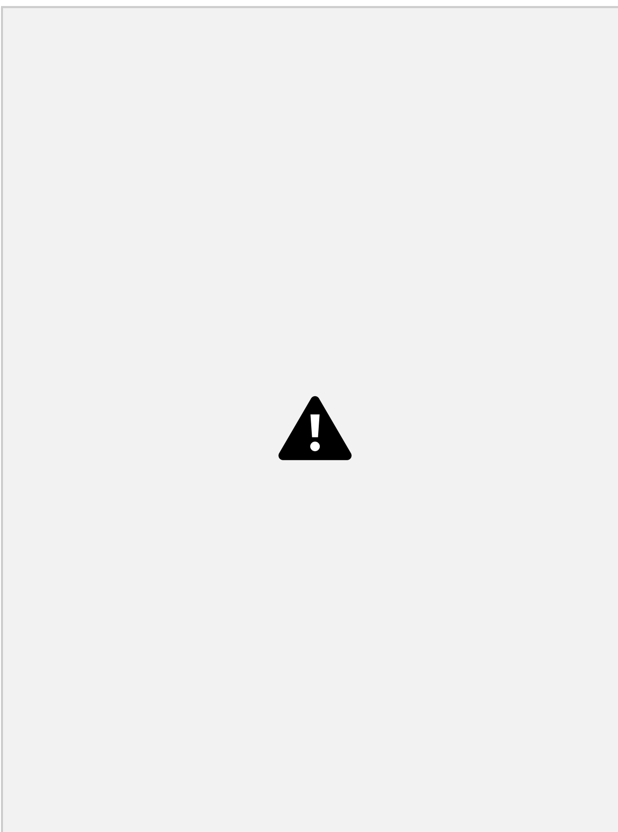
click done and continue

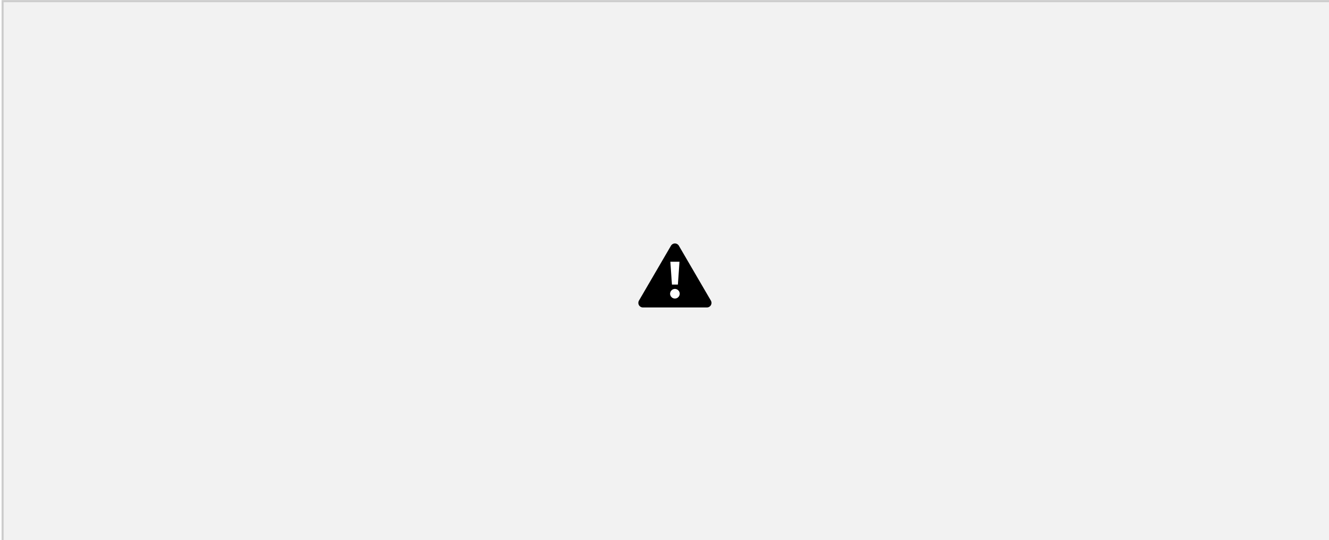






Wait to finish





After the finish log in and put in this

- sudo systemctl status ssh – for checking the SSH status.
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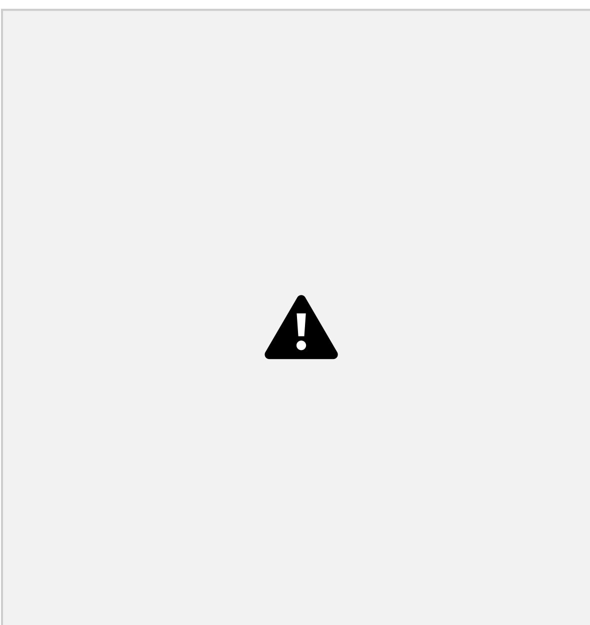

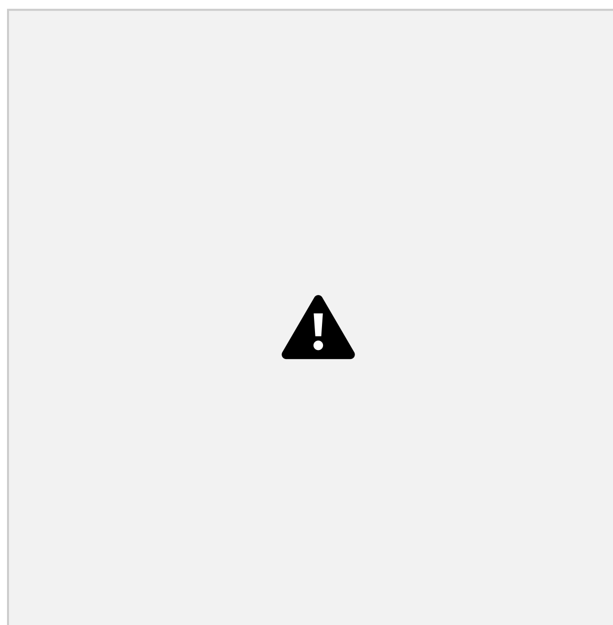

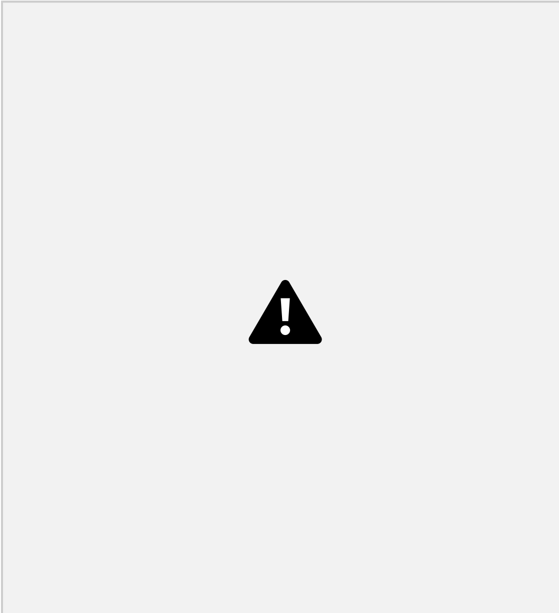 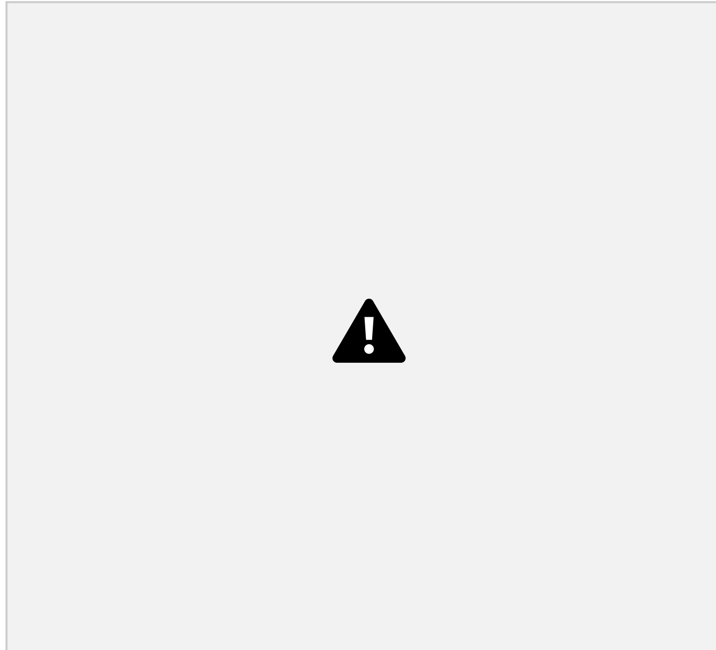

After that log in again and put a ip a and sudo dhcpcd, enter a password and successfull